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Bihar Engineering University, Patna

B.Tech 2nd Semester Examination, 2025 (Old Course)

Time: 03 Hours

Code: 105201 ***Subject: Physics (Semiconductor Physics)*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option/answer the following (Any seven question only): **[14]**

(a) Source for Optical fibre communication can be
(i) LED
(ii) Ordinary light
(iii) LASER
(iv) Both (i) & (ii)

(b) A light emitting diode is
(i) Lightly doped
(ii) Heavily doped
(iii) Zener Diode
(iv) Intrinsic Semiconductor

(c) What are phonons?

(d) Give an example of indirect bandgap semiconductor.

(e) Fermi level of Potassium is 2.1 eV. Calculate the speed of electron at Fermi level. (mass of electron = 9.1×10^{-31}kg)

(f) Write down any two characteristics of wave function.

(g) Write down time-independent Schrödinger's wave equation in 1-dimension.

(h) The de Broglie wavelength of a particle moving with a velocity 200 m/s is 14.14 Å. What will be de Broglie wavelength if kinetic energy is doubled?

(i) State Heisenberg's uncertainty principle.

(j) Define mobility of a charge carrier. Give its S.I. Unit.

Q.2 Solve both questions : **[14]**

(a) Discuss Kronig-Penney model for the motion of an electron in a periodic potential and the existence of allowed and forbidden energy bands.

(b) Find out the energy level in Sodium for which the probability of occupation at 300K is 0.50. (Given Fermi energy for Sodium is 3.13 eV. Planck's constant = 1.38×10^{-23} in S.I. Units)

Q.3 Solve both questions : **[14]**

(a) Discuss the structure and working of semiconductor LASER. Draw and explain its energy band diagram.

(b) What is population inversion? What is its importance in Lasing action?

Q.4 Solve both questions : **[14]**

(a) Derive an expression for effective mass of an electron.

(b) Explain the significance of negative effective mass.

Q.5 Write short notes on : **[14]**

(a) Advantages and Disadvantages of solar cell

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(b) Photodetectors and its application

Q.6 Solve both questions : **[14]**

(a) Derive the expression for the equilibrium concentration of electrons in the conduction band.

(b) Explain the effect of temperature on Fermi level.

Q.7 Write notes on any two of the following: **[14]**

(a) Pumping mechanisms in LASER

(b) Bloch Theorem

(c) Dependence of Fermi energy on carrier concentration

Q.8 Solve this question : **[14]**

What are LEDs? Sketch the relative eye response as a function of wavelength. Describe device structure and materials used for LED with its characteristics.

Q.9 Solve both questions : **[14]**

(a) Discuss the structure, working principle and characteristics of PIN and Avalanche Photodiodes.

(b) Explain the effect of temperature on Fermi level.

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